Rethinking Efficiency Bottleneck of Generative Robotics Manipulation

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Biography

Academia

Industry

2026

2025

2024

2023

2022

Sep 2025 – Jan 2024
Microsoft Research Asia
StarTrack Scientist
Image Compression with Agents

May 2024 – Oct 2024
TikTok Australia
Research Engineer
Trust and Safety
Generative AI with LLM

Oct 2024 - Now
The University of Newcastle
Lecturer (Continuing)

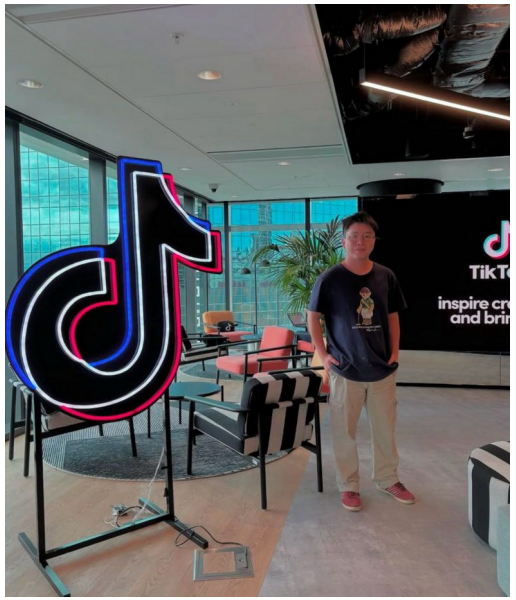
Sep 2022 - May 2024
The University of Sydney
Postdoctoral Fellow

Sep 2022 - Feb 2023
Disney Research Switzerland
Visiting Scientist
AI-based Video Compression System with Deep Learning

Mar 2019 – Dec 2022
The University of Sydney
B.ENG (2017) & Ph.D. (2022)



Oct 2022
Disney Research
Zurich Switzerland



Sep 2024
TikTok
Sydney Australia



Dec 2025
Microsoft Research
Beijing China

Generative Robotics Manipulation

- Principle
- Methods
- Issue



Application of Generative AI

Automatic Robotics Manipulations

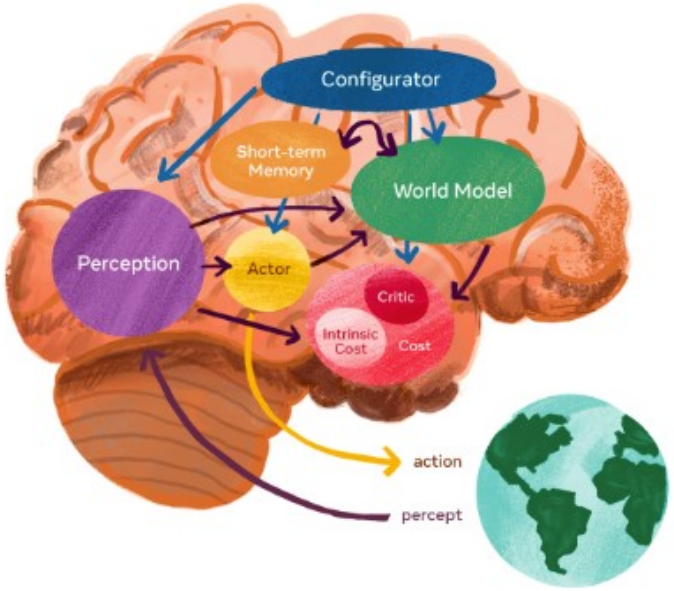
Generative models produce robot actions from visual observations, robot state and a task goal.

"Put the red mug on the tray."











An approach to Embodied AI

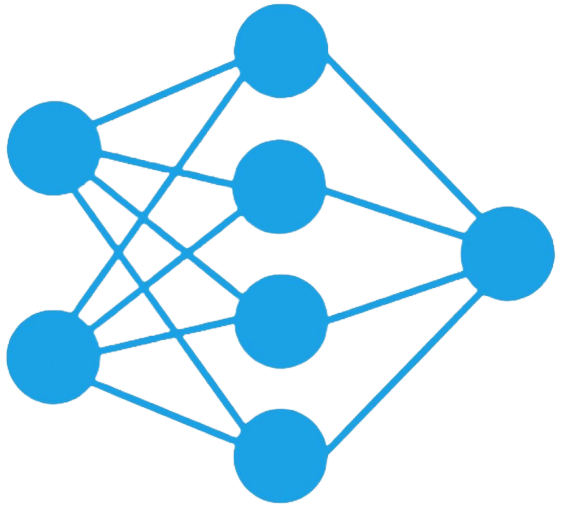
Intelligence grounded in physical interaction through a perception–action feedback loop.

Vision Language Action (VLA) Model
World Action Model
Combine powerful Vision Language Model and Diffusion Policy
Leverage a powerful world model

Generative AI: Scaling Law

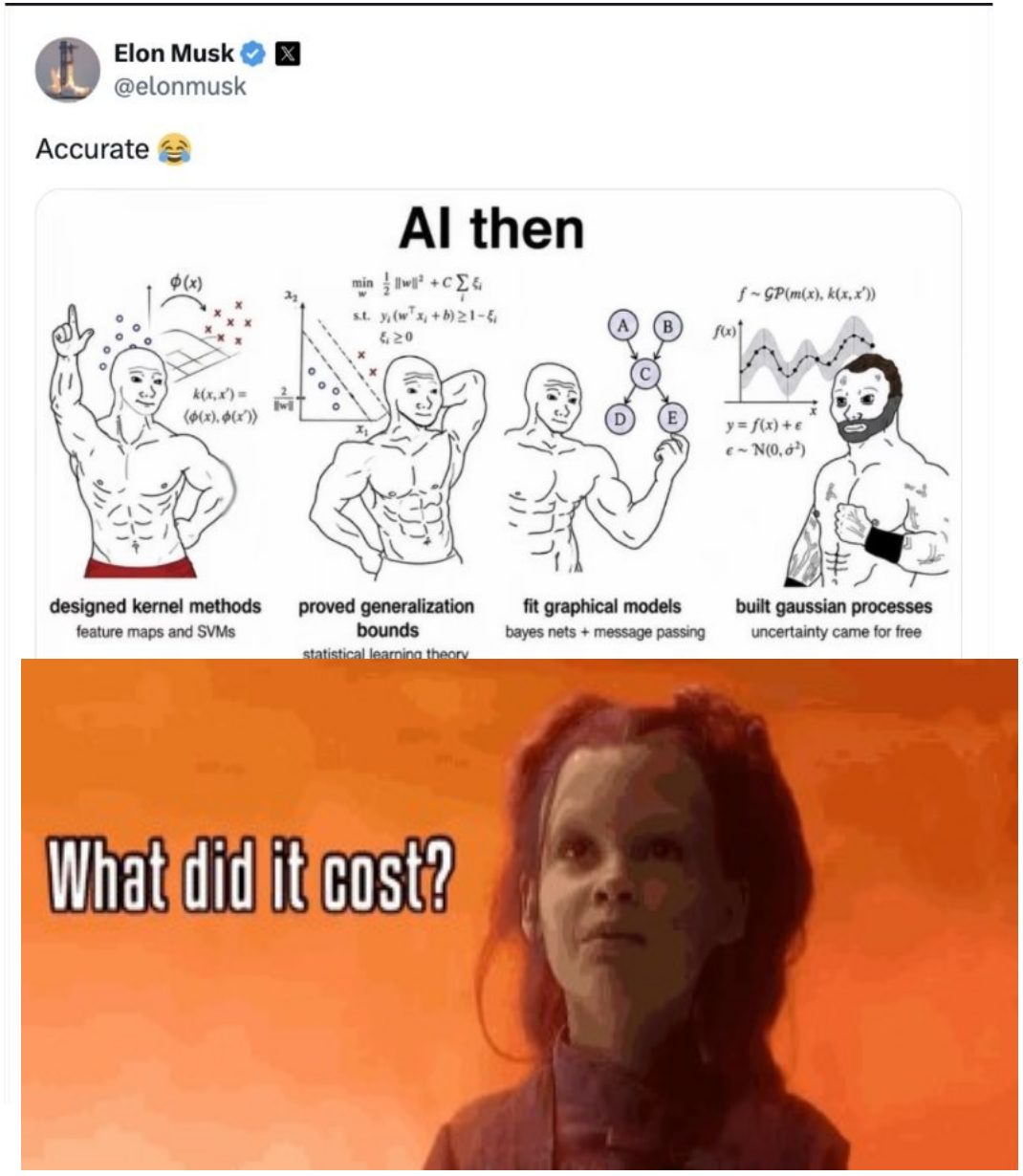






Large Scale Data

Large Scale Model



Data Efficiency

- Robotic Training Data
- Compression Foundation
- 3DGS Compression
- Real-time 4DGS Compression
- Real-time 4DGS Compression and Editting
- A Public Dataset



Robotics Data –
Multiview 3D/4D Scenes

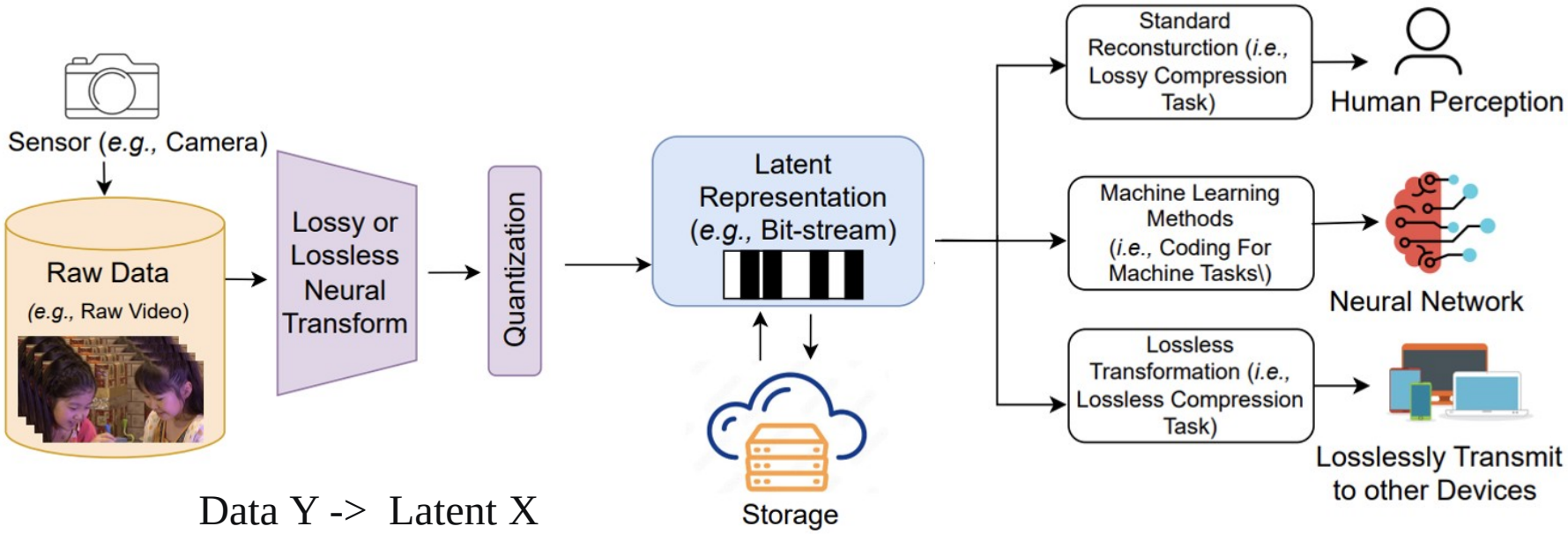


Collection

Rendering

A Generative Model Solution for Data Compression

Generative Model (*e.g.*, VAE) to **estimate** the ***distribution* of X, *q(X)*** so the cross entropy (*i.e.*, energy) can be smaller

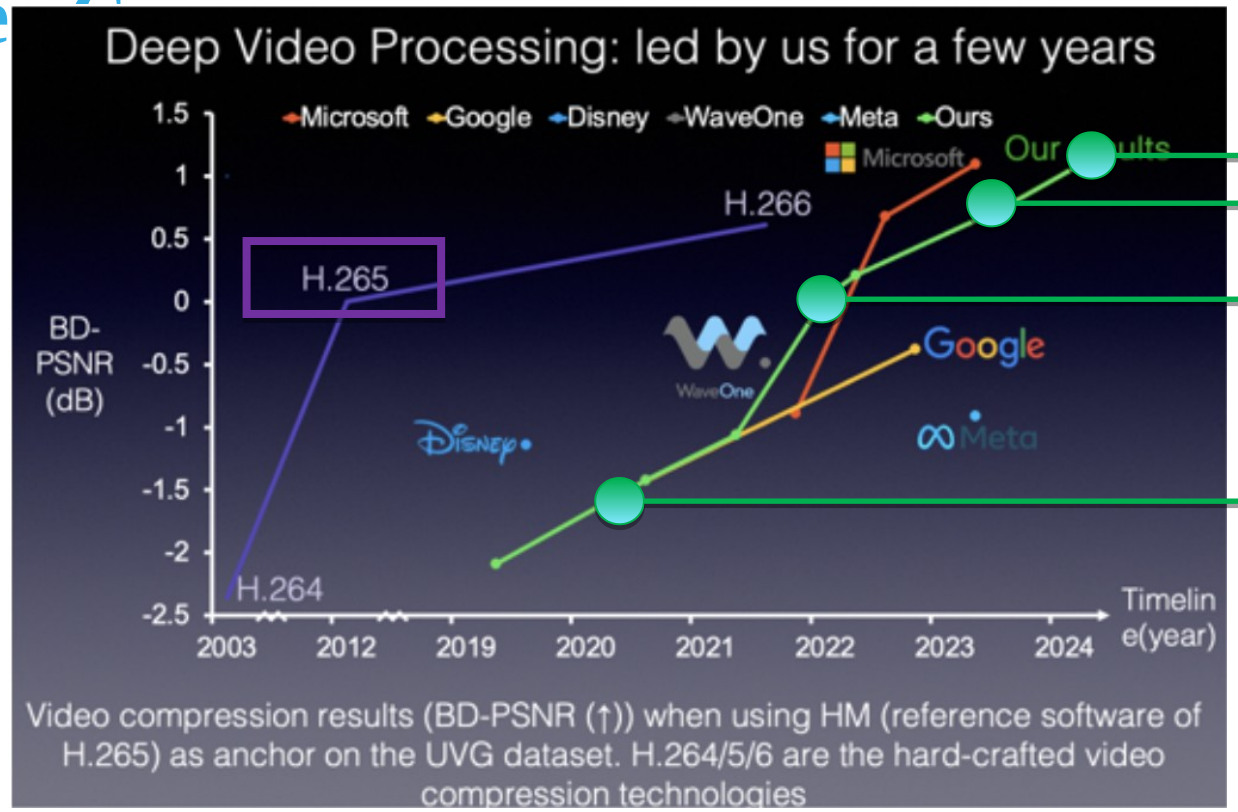


Data Y -> Latent X

$$H(p,q) = -\sum_{x \in \mathcal{X}} \underbrace{p(x)}_{\text{real}} \log \underbrace{q(x)}_{\text{estimated}} \quad \text{(Eq.1)}$$

Lossy Compression – 2D Video C

We are leading at the Video Compression Task for years



***Chen et al.*, MM2024**

***Chen et al.*, MM2023 / US Patent 2024**

***Chen et al.*, CVPR2022**

***Hu & Chen et al.*, ECCV2020**

[Chen et al, MM2024]. "Group-aware Parameter-efficient Updating for Content-Adaptive Neural Video Compression." in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024
[Chen et al, US Patent 24] "Contextual video compression framework with spatial-temporal cross-covariance Transformer" in US Patent 18/349,076
[Chen et al, MM2023]. "Neural Video Compression with Spatio-Temporal Cross-Covariance Transformers." in *Proceedings of the 31st ACM International Conference on Multimedia*, 2023
[Chen et al, CVPR2022] "LSVC: A learning-based stereo video compression framework", in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022*
[Hu & Chen et al, ECCV2020] "Improving deep video compression by resolution-adaptive flow coding" in *European Conference on Computer Vision, 2020*

II. 3D Scenes Compression

- 3D Gaussian Splatting (3DGS) support the Real-Time Rendering is the trend for 3D Game

[ArXiv'2025] L Liu, Z Chen, W Jiang, W Wang, D Xu, *HEMGS: A Hybrid Entropy Model for 3D Gaussian Splatting Data Compression*, arXiv: 2411.18473

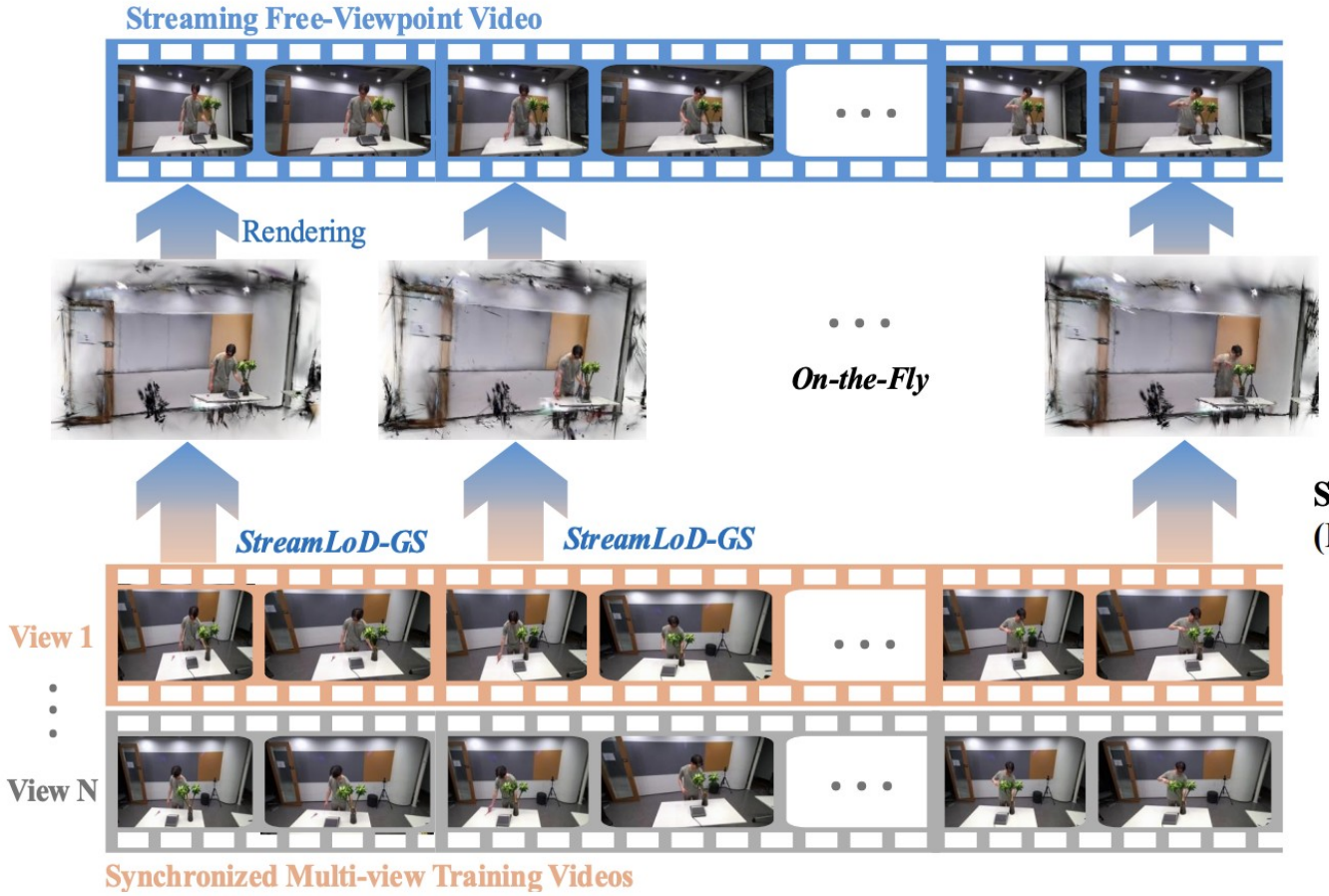
Method	Rank	Tanks and Temples				Mip-NeRF 360				Deep Blending				Synthetic NeRF			
		PSNR	SSIM	LPIPS	Size [MB]	PSNR	SSIM	LPIPS	Size [MB]	PSNR	SSIM	LPIPS	Size [MB]	PSNR	SSIM	LPIPS	Size [MB]
HEMGS-highrate	5.7	24.58	0.856	0.176	10.1	27.93	0.813	0.230	21.0	30.37	0.911	0.253	6.7	33.71	0.968	0.035	1.6
HEMGS-lowrate	6.2	24.42	0.848	0.192	6.0	27.75	0.806	0.248	12.5	30.24	0.908	0.266	3.0	33.58	0.967	0.037	1.3
HAC-highrate	7.1	24.40	0.853	0.177	11.8	27.77	0.811	0.230	22.9	30.34	0.906	0.258	6.7	33.71	0.968	0.034	2.0
HAC++-highrate	5.1	24.33	0.853	0.181	7.3	27.82	0.811	0.231	19.4	30.34	0.911	0.254	5.5	33.76	0.969	0.033	1.9
ContextGS_highrate	8.3	24.29	0.855	0.176	12.4	27.75	0.811	0.231	19.3	30.41	0.909	0.259	6.9				
HAC++-lowrate	7.1	24.22	0.849	0.190	5.4	27.60	0.803	0.253	8.7	30.16	0.907	0.266	3.1	32.77	0.965	0.041	0.8
ContextGS_lowrate	7.1	24.12	0.849	0.186	9.9	27.62	0.808	0.237	13.3	30.09	0.907	0.265	3.7				
HAC-lowrate	7.7	24.04	0.846	0.187	8.5	27.53	0.807	0.238	16.0	29.98	0.902	0.269	4.6	33.24	0.967	0.037	1.2
gsplat-1.00M	7.8	24.03	0.857	0.163	16.1	27.29	0.811	0.229	16.0								
Scaffold-GS	19.0	23.96	0.853	0.177	87.0	27.50	0.806	0.252	156.0	30.21	0.906	0.254	66.0				
CompGS	13.7	23.70	0.837	0.208	10.1	27.26	0.803	0.239	17.3	29.69	0.901	0.279	9.2				
CodecGS	7.9	23.63	0.841	0.192	7.8	27.30	0.810	0.236	10.3	29.81	0.906	0.251	9.0				
FCGS-highrate	18.8	23.62	0.839	0.184	33.6	27.39	0.806	0.226	67.2	29.58	0.899	0.248	54.5				
Reduced3DGS	14.0	23.57	0.840	0.188	14.0	27.10	0.809	0.226	29.0	29.63	0.902	0.249	18.0				
SOG	17.0	23.56	0.837	0.186	22.8	27.08	0.799	0.230	40.3	29.26	0.894	0.268	17.7	33.23	0.966	0.034	4.1

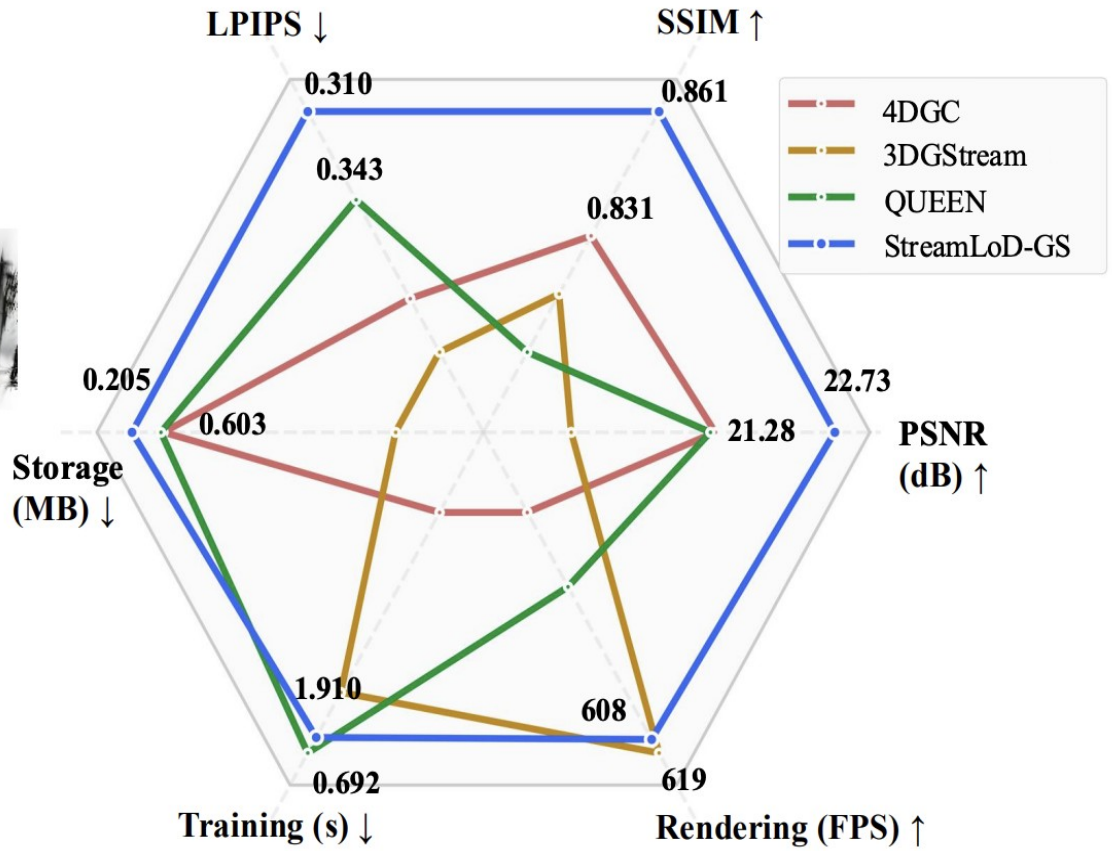
https://w-m.github.io/3dgs-compression-survey/

[ArXiv'2025] L Liu, Z Chen, W Jiang, W Wang, D Xu *HEMGS: A Hybrid Entropy Model for 3D Gaussian Splatting Data Compression,* arXiv: 2411.18473

III. Real-time 4D Scene

It also support Real Time Immersive Streaming

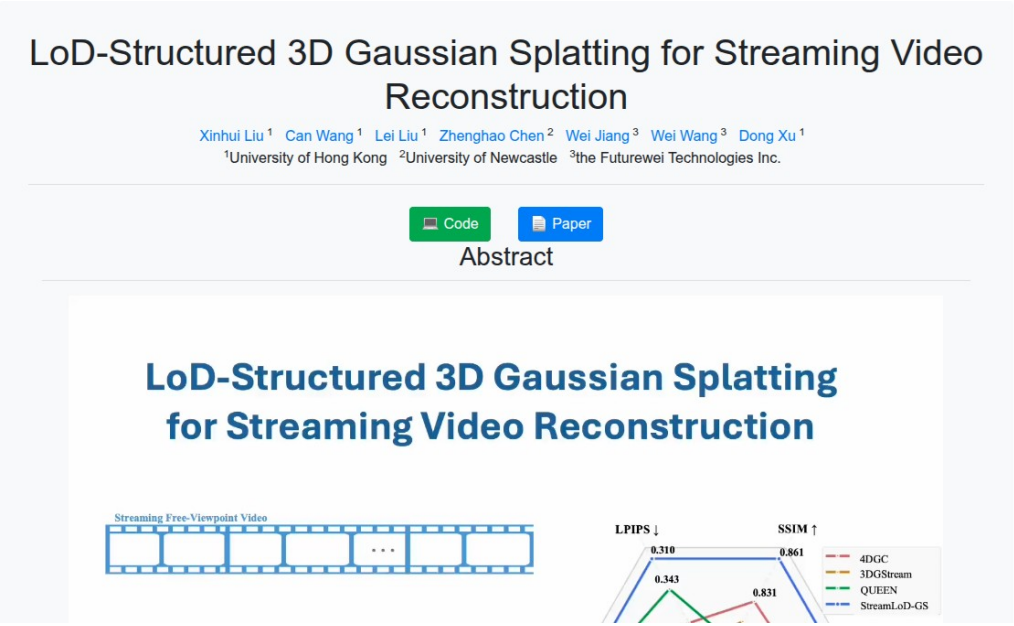


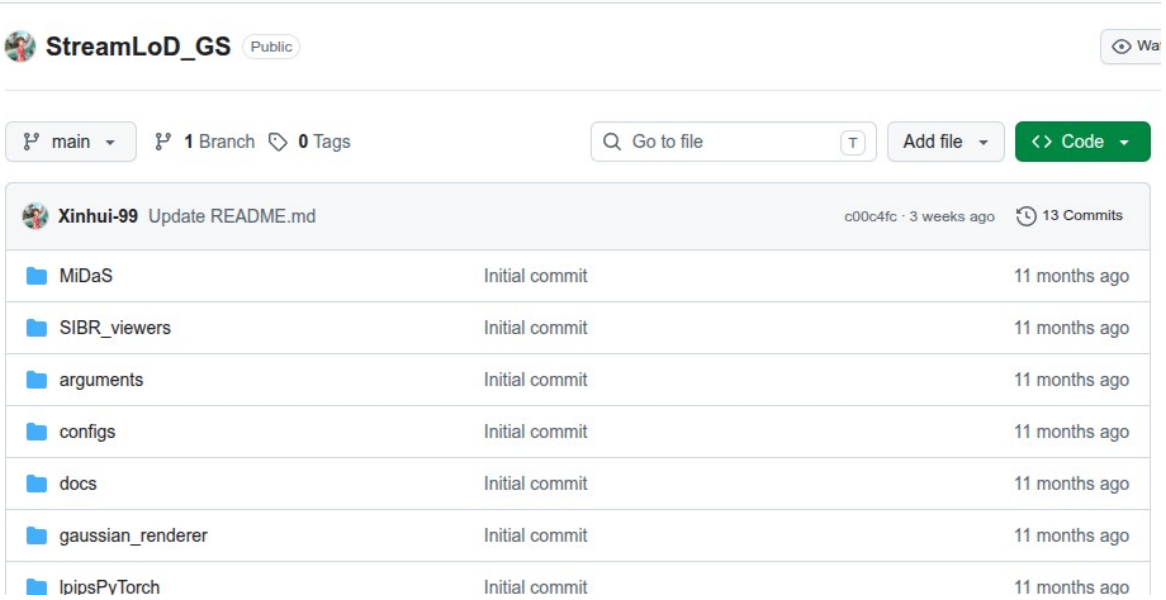


[IEEE TVCG 2027] "LoD-Structured 3D Gaussian Splatting for Streaming Video Reconstruction", X Liu, C Wang, L Liu, Z Chen, W Jiang, W Wang, D Xu, *IEEE Transactions on Visualization and Computer Graphics 2027*

III. Real-time 4D Scene Compression

https://xinhui-99.github.io/StreamLoD_GS_progect_page/

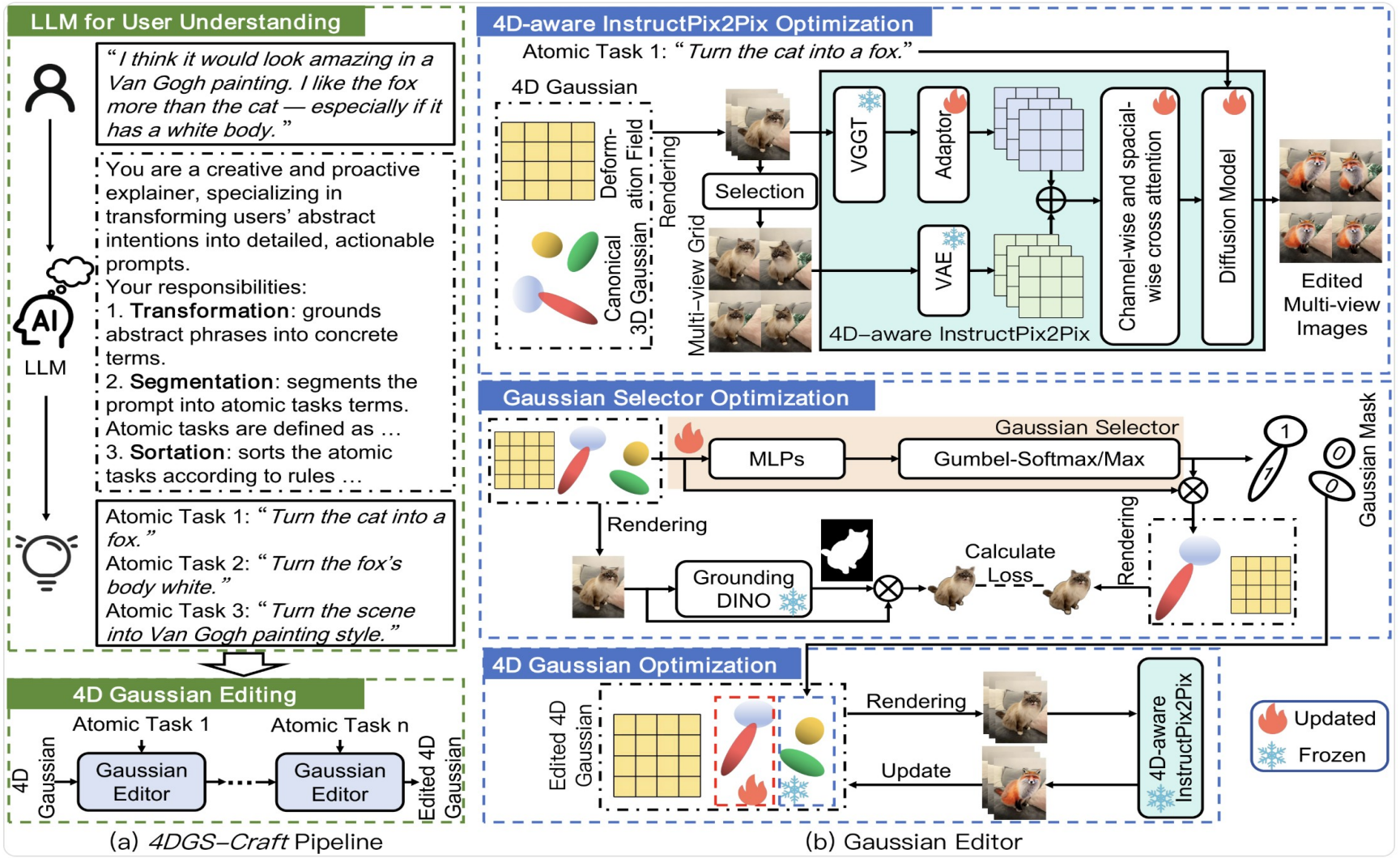




[IEEE TVCG 2027] "LoD-Structured 3D Gaussian Splatting for Streaming Video Reconstruction", X Liu, C Wang, L Liu, Z Chen, W Jiang, W Wang, D Xu, *IEEE Transactions on Visualization and Computer Graphics 2027*

IV. 4] Com time

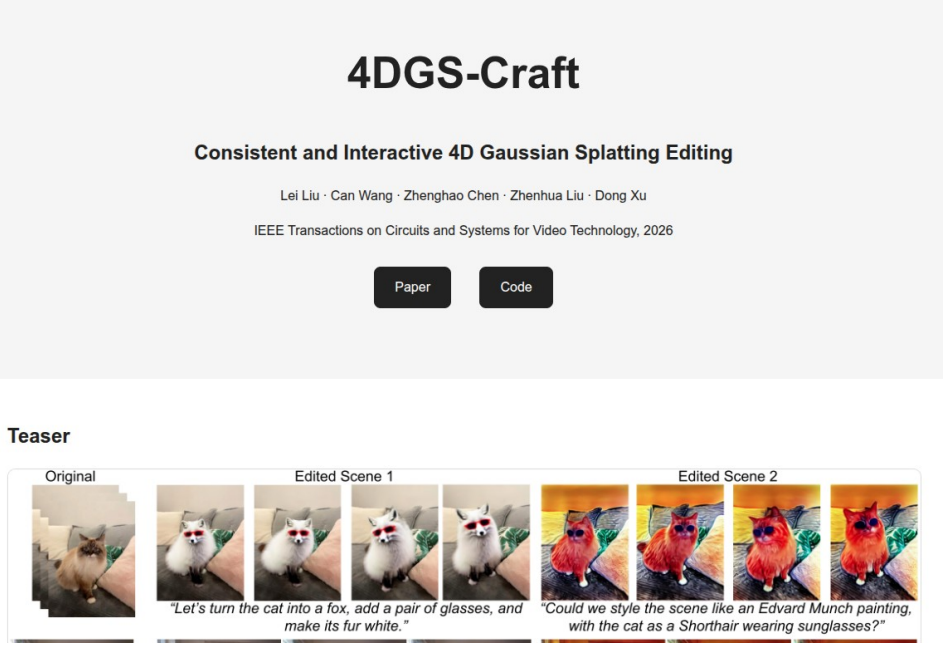
You can directly edit 3D or 4D Gaussians to produce new re-styled multi-view videos.

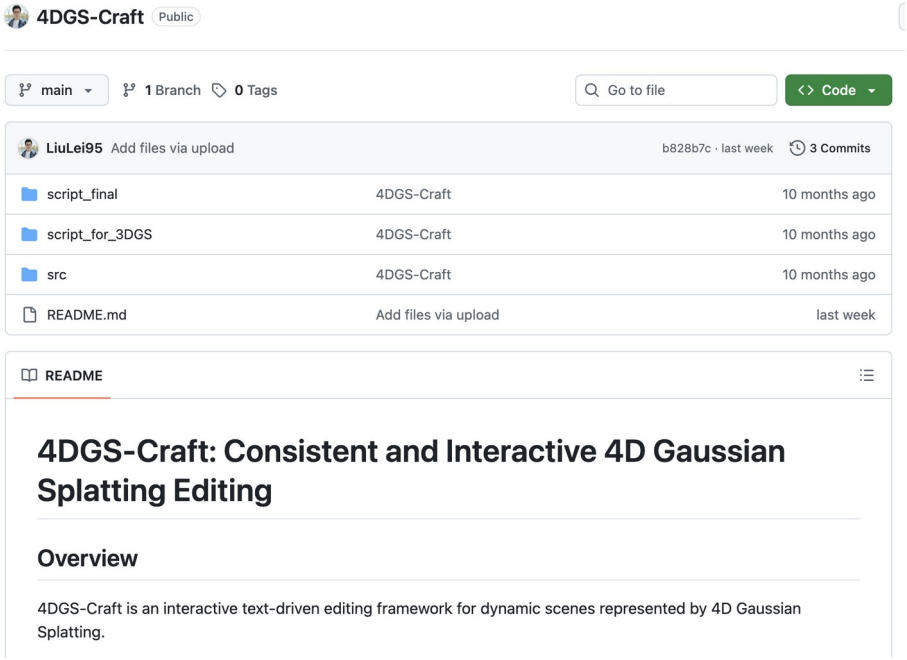


[IEEE T-CSVT'2026] Liu, L., Wang, C, Chen, Z., & Xu, D. (2026). "4DGS-Craft: Consistent and Interactive 4D Gaussian Splatting Editing" , *in IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, 2026*

IV. 4D Scene Compression with Real-time Editing

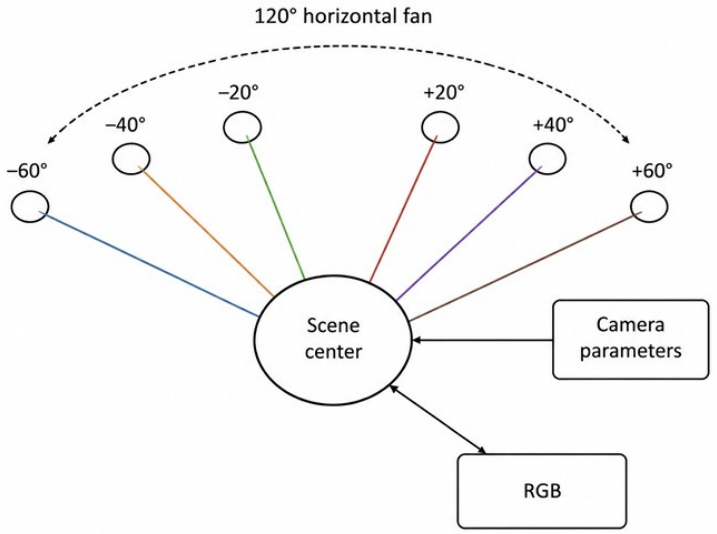
https://liulei95.github.io/4DGS-Craft-ProjectPage/

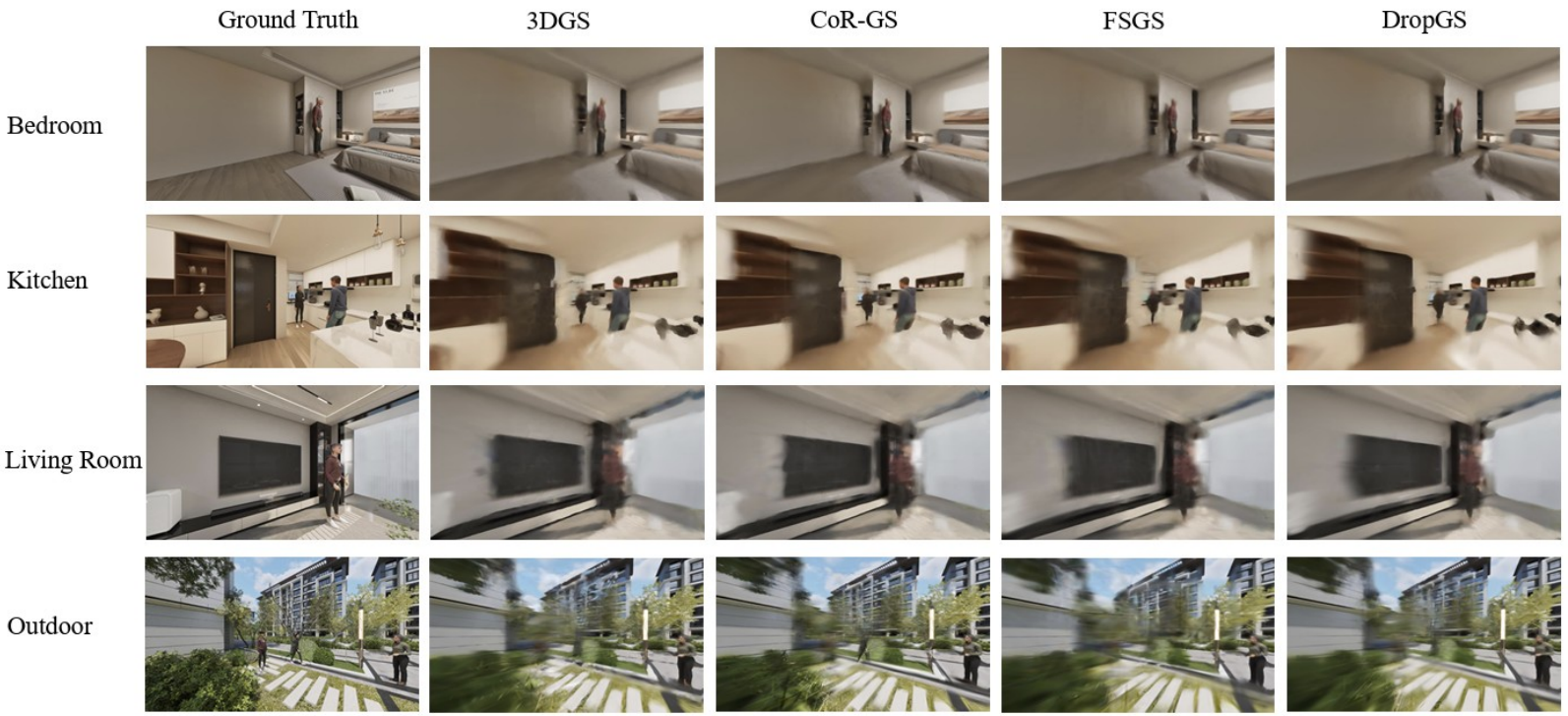




[IEEE T-CSVT'2026] Liu, L., Wang, C, Chen, Z., & Xu, D. (2026). "4DGS-Craft: Consistent and Interactive 4D Gaussian Splatting Editing" , *in IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, 2026*

V. A Public Dataset fo





Method	PSNR ↑	SSIM ↑	LPIPS ↓	Storage (MB) ↓
3DGS	18.705	0.655	0.461	1331
CoR-GS	18.825	**0.658**	**0.445**	3174
FSGS	18.726	0.654	0.475	**508**
DropGaussian	**18.864**	0.657	0.466	992

Method	PSNR ↑	SSIM ↑	LPIPS ↓	Storage (MB) ↓	Train/Recon. Time (s) ↓	Render FPS ↑
QUEEN [14]	15.93	0.5125	0.4786	1734	**351.48**	54.64
Octree-/LoD-structured [31]	**16.32**	**0.5336**	**0.4650**	**1452**	356.90	**60.10**

Method	PSNR ↑	SSIM ↑	LPIPS ↓	Storage (MB) ↓	FPS ↑
Deformable 3DGS [51]	**22.79**	0.7258	0.4024	257	12.84
4DGS [47]	22.57	**0.7323**	**0.3598**	**200**	**52.84**

[MM'2026] Liu, Liu and Chen et al. (2026). "MISR: A Multi-Modal Multi-View Benchmark for 3D/4D Gaussian Splatting and Feedforward Compression" *in ACM MM2026*

https://richmediagai.github.io/



4th RichMediaGAI ACM Multimedia 2026

ACM Multimedia 2026
Rio de Janeiro, Brazil

4th International Workshop on Rich Media with Generative AI

Workshop Homepage: https://richmediagai.github.io
Paper Submission Site: https://openreview.net/group?id=acmmm.org/ACMMM/2026/Workshop/RichMediaGAI
Accepted papers will be published in proceedings by Sheridan Publishing on behalf of ACM

Call For Papers

The **RichMediaGAI2026** workshop focuses on the intersection of generative AI and rich media systems, aiming to bridge content creation, representation, compression, and delivery within unified end-to-end pipelines. The scope spans visual, multimodal, and 3D/4D media, addressing both algorithmic advances in generative modeling and system-level challenges related to scalability, efficiency, and real-world deployment.

A key emphasis of **RichMediaGAI2026** is free-view video (FVV) as a representative and demanding use case for generative media systems. FVV highlights the need to jointly consider scene representation, temporal consistency, compression, interactive delivery, going beyond reconstruction quality alone. To support reproducible evaluation, the workshop is accompanied by the **RichMediaGAI Challenge 2026**, built on the **M3VIR-2** benchmark, which provides a common testbed for assessing generative synthesis and delivery efficiency under realistic constraints. While FVV serves as a central theme, the workshop is not limited to FVV and broadly covers generative techniques for images, videos, and immersive media applications. The **RichMediaGAI2026** workshop aims to foster systematic comparison, community engagement, and progress toward deployable generative media systems.

Interested Topics

Generative FVV
- 3D/4D scene synthesis
- 3D/4D Gaussian Splatting

Compression and Streaming
- Compression and streaming of neural/Gaussian scene
- Spatiotemporal-consistent video generation

Real-Time Interaction
- Real-time neural rendering
- Interactive free-view navigation

Benchmark
- Dataset and benchmark for FVV generation/delivery

Evaluation
- Metric for view consistency, temporal stability
- Bitrate/latency, QoE

Applications
- AR/VR
- Robotics/Embodied AI
- Digital twins

Important Dates

- Regular paper submission deadline: July 26, 2025
- Challenge paper submission deadline: August 9, 2025
- Challenge result submission deadline: August 9, 2025
- Notification of acceptance: August 12, 2025
- Camera-ready submission deadline: August 20, 2026

Call for Challenge Participants

The workshop features a challenge focused on generative FVV synthesis and efficient neural scene delivery. It is built on the **M3VIR-2** benchmark, which provides scenes containing static environments and dynamic objects rendered from synchronized egocentric cameras, along with ground-truth depth, semantic and instance segmentation, and static-dynamic masks. The challenge tracks include topics related to 3D/4D Gaussian Splatting and compression. Top-performing participants will be invited to submit papers.

Invited Speakers

Junsong Yuan
Professor
University at Buffalo SUNY

Jinwei Gu
Senior Manager & Principal Research Scientist
Adjunct Professor
Nvidia, CUHK

Rogerio Feris
Senior Research Manager & Principal Scientist
MIT-IBM Watson AI Lab

Mike Zheng Shou
Assistant Professor
National University of Singapore

Organizers

Wei Jiang
Senior Principal Researcher
Futurewei Technologies

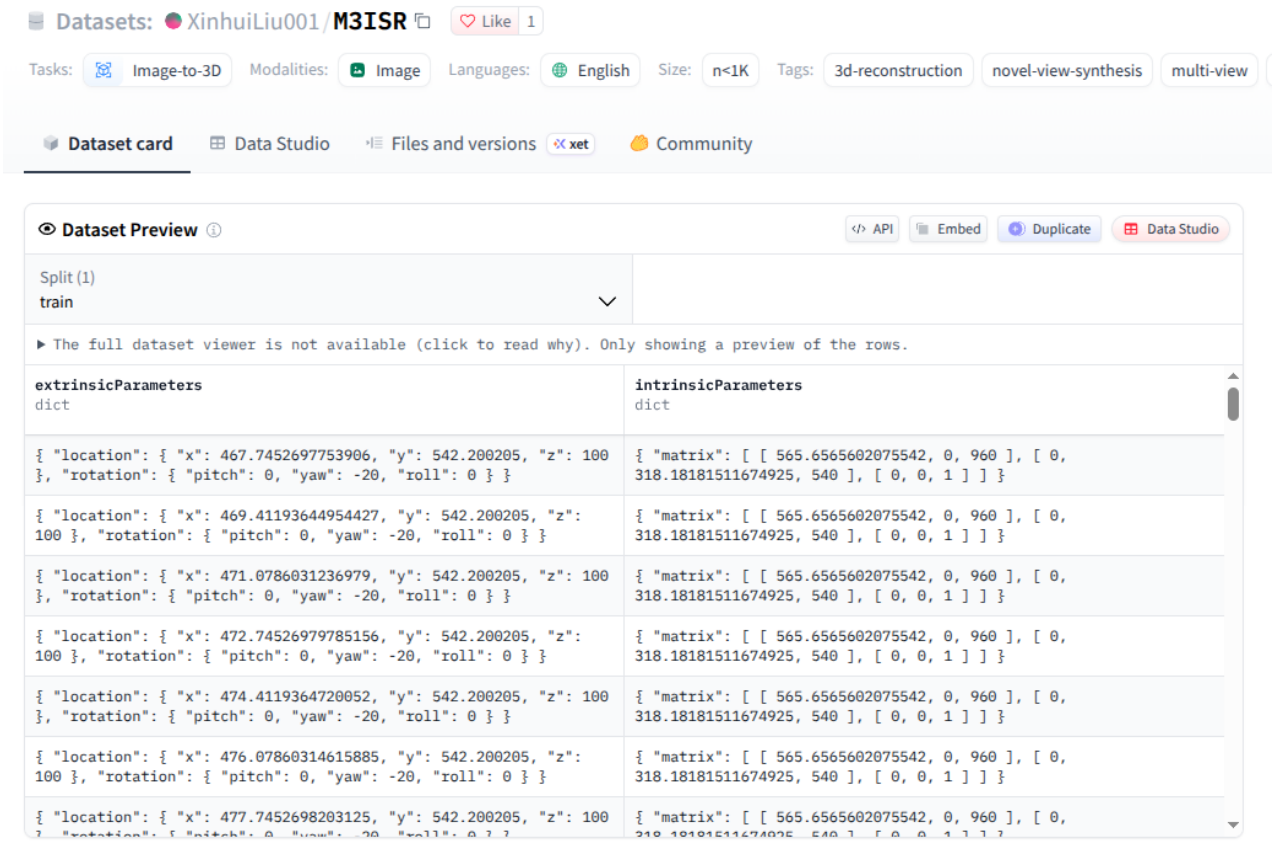
Dong Xu
Professor
University of Hong Kong

Zhenghao Chen
Assistant Professor
University of NewCastle

FUTUREWEI

THE UNIVERSITY OF NEWCASTLE AUSTRALIA

THE UNIVERSITY OF HONG KONG



Datasets: XinhuiLiu001 / **M3ISR** Like 1

Tasks: Image-to-3D | Modalities: Image | Languages: English | Size: n<1K | Tags: 3d-reconstruction | novel-view-synthesis | multi-view

Dataset card | Data Studio | Files and versions | xet | Community

Dataset Preview API | Embed | Duplicate | Data Studio

Split (1)
train

▶ The full dataset viewer is not available (click to read why). Only showing a preview of the rows.

extrinsicParameters dict	intrinsicParameters dict
{ "location": { "x": 467.7452697753906, "y": 542.200205, "z": 100 }, "rotation": { "pitch": 0, "yaw": -20, "roll": 0 } }	{ "matrix": [[565.6565602075542, 0, 960], [0, 318.18181511674925, 540], [0, 0, 1]] }
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{ "location": { "x": 476.07860314615885, "y": 542.200205, "z": 100 }, "rotation": { "pitch": 0, "yaw": -20, "roll": 0 } }	{ "matrix": [[565.6565602075542, 0, 960], [0, 318.18181511674925, 540], [0, 0, 1]] }
{ "location": { "x": 477.7452698203125, "y": 542.200205, "z": 100	{ "matrix": [[565.6565602075542, 0, 960], [0,

[MM'2026] Liu, Liu and Chen et al. (2026). "MISR: A Multi-Modal Multi-View Benchmark for 3D/4D Gaussian Splatting and Feedforward Compression" *in ACM MM2026*

Model Efficiency

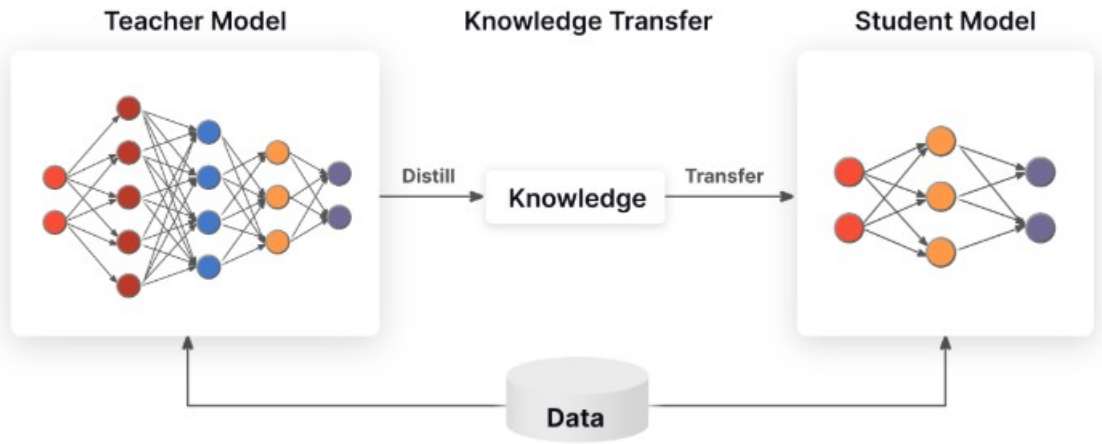
- **Problem Formulation**
- **Large Language Model**
- **Robotics Embodied AI Model**
  - –
-



Model Efficiency

Problem Models are often large and slow, making them hard to deploy on edge devices or in real-time applications.

Methods Model compression reduces size and computation without significantly sacrificing accuracy. Common techniques include:
– *Pruning* (removing redundant weights)
– *Quantization* (using lower precision)
– *Knowledge Distillation* (training a small model to mimic a large one)



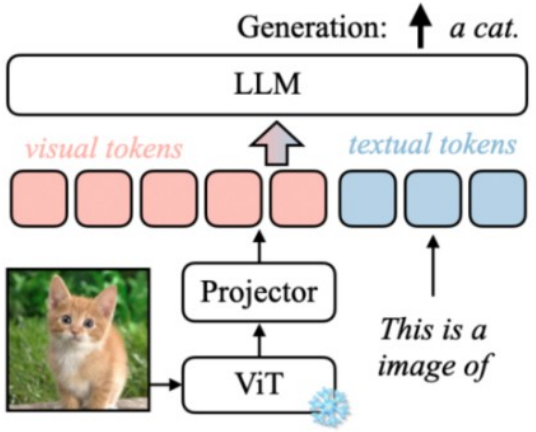
Formulation

Given a large model M, find a compressed model M' such that:

$$\text{Size}(M') \ll \text{Size}(M), \quad \text{Accuracy}(M') \approx \text{Accuracy}(M)$$

Video Large Language Model Efficiency

- Text-GenAI Model (e.g., LLM), texts is Compact and Informative.
- Video GenAI Model (e.g., VLM, T2V), visual signal is Redundant and Antonym

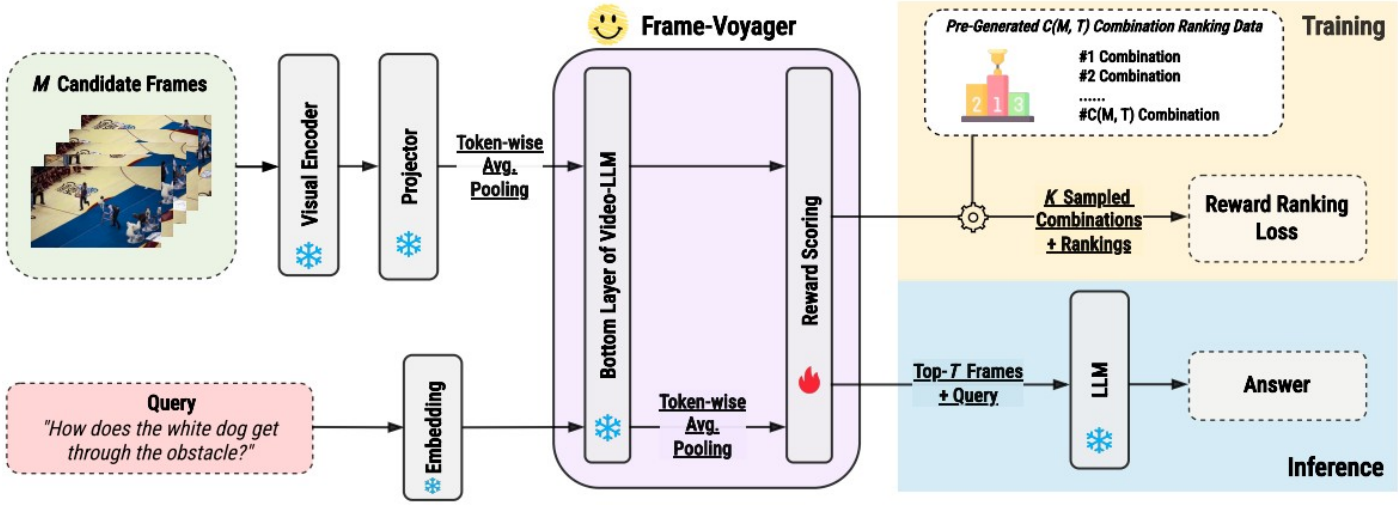


Text Tokens < 10

Visual Tokens > 100 (e.g., 576 in LLAVA (CLIP/ViT) for per image)

A 5 min 60 FPS video
5 x 60 x 60 x 576
> 10 Millions

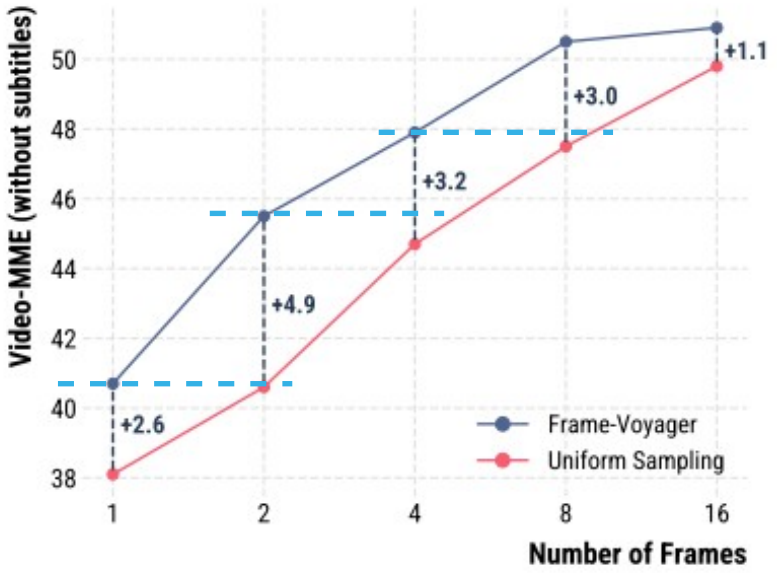
We carefully select Key Frames (aka. Key Visual Tokens)!
[*Yu, Jin, Wang & Chen et al*, ICLR2025]



[ICLR2025] *S Yu, C Jin, H Wang & Z Chen et al, "Frame-Voyager: Learning to Query Frames for Video Large Language Models" in* The International Conference on Learning Representations 2025

Address the Redundant Visual Features in Video-LLM

Model	LLM Size	Video-MME (w/o subs / w subs)				MLVU	ANQA	NextQA
Video Length		Overall *17min*	Short *1.3min*	Medium *9min*	Long *41min*	*12min*	*2min*	*0.8min*
Video-LLaVA	7B	39.9 / 41.6	45.3 / 46.1	38.0 / 40.7	36.2 / 38.1	47.3	45.3	-
Qwen-VL-Chat	7B	41.1 / 41.9	46.9 / 47.3	38.7 / 40.4	37.8 / 37.9	-	-	-
ST-LLM	7B	37.9 / 42.3	45.7 / 48.4	36.8 / 41.4	31.3 / 36.9	-	50.9	-
VideoChat2	7B	39.5 / 43.8	48.3 / 52.8	37.0 / 39.4	33.2 / 39.2	44.5	49.1	-
Chat-UniVi-V1.5	7B	40.6 / 45.9	45.7 / 51.2	40.3 / 44.6	35.8 / 41.8	-	46.1	-
VideoLLaMA2	7B	47.9 / -	56.0 / -	45.4 / -	42.1 / -	-	49.9	-
LLaVA-NeXT-QW2	7B	49.5 / -	58.0 / -	47.0 / -	43.4 / -	-	-	-
LongVILA^{128frm}	8B	49.2 / -	60.2 / -	48.2 / -	38.8 / -	-	-	-
LongVILA^{256frm}	8B	50.5 / -	**61.8** / -	**49.7** / -	39.7 / -	-	-	-
VILA*	8B	47.5 / 50.0	57.8 / 61.6	44.3 / 46.2	40.3 / 42.1	46.3	44.3	48.5
+FRAME-VOYAGER	8B	**50.5 / 53.6**	60.3 / **65.0**	47.3 / **50.3**	**43.9 / 45.3**	**49.8**	**51.1**	**51.6**
VideoLLaMA2	8×7B	47.9 / 49.7	- / -	- / -	- / -	-	50.3	-
VITA	8×7B	55.8 / 59.2	65.9 / 70.4	52.9 / 56.2	48.6 / 50.9	-	-	-
LLaVA-NeXT-Video	34B	52.0 / 54.9	61.7 / 65.1	50.1 / 52.2	44.3 / 47.2	-	**58.8**	-
VILA*	34B	58.3 / 61.6	67.9 / 70.7	56.4 / 59.8	50.4 / 52.1	57.8	47.7	67.4
+FRAME-VOYAGER	34B	**60.0 / 63.8**	**70.3 / 73.1**	**58.3 / 62.7**	**51.2 / 55.7**	**61.1**	55.2	**69.4**

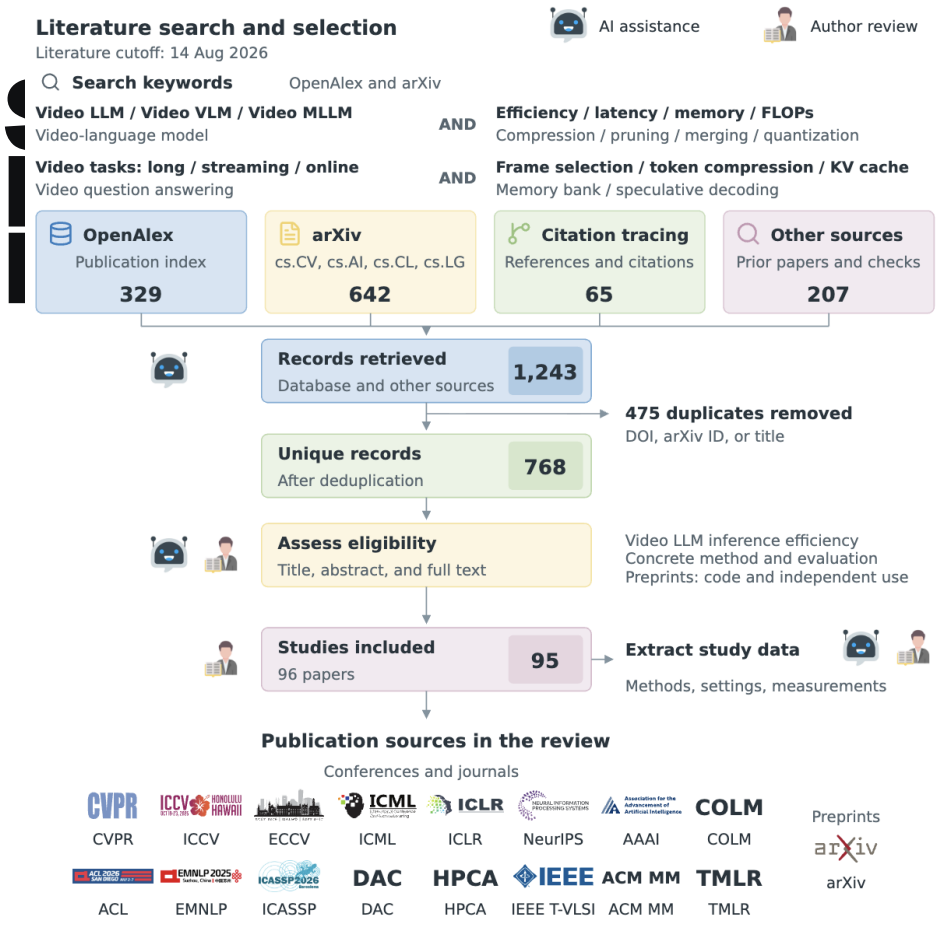


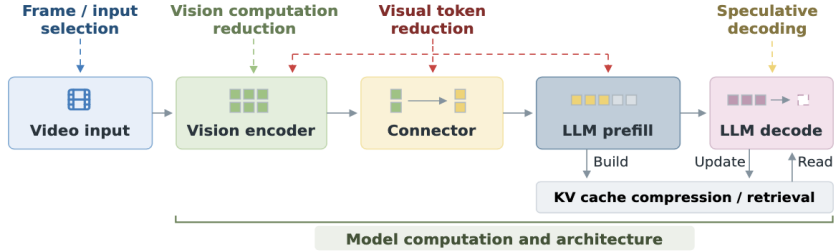
- Simply increasing the visual information (i.e., visual tokens) does not help your LLM!

-

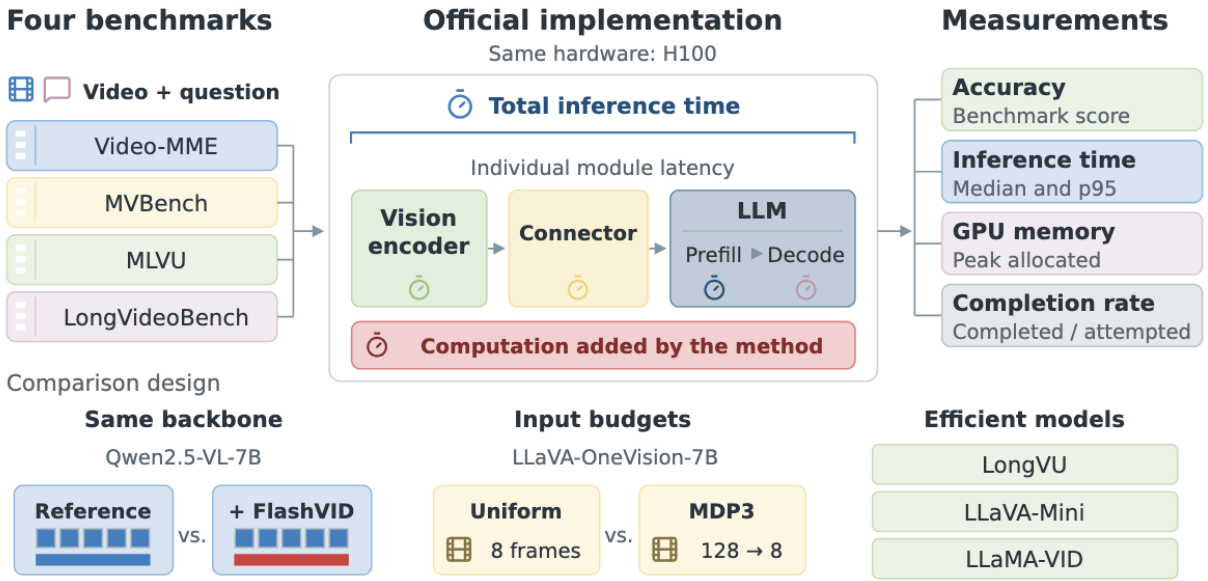
- The key is to extract the compact and useful tokens (**aka Model Efficiency**!)

[ICLR2025] *S Yu, C Jin, H Wang & Z Chen et al, "Frame-Voyager: Learning to Query Frames for Video Large Language Models" in* The International Conference on Learning Representations 2025





Identify 8 major categories of Efficiency Video LLM methods



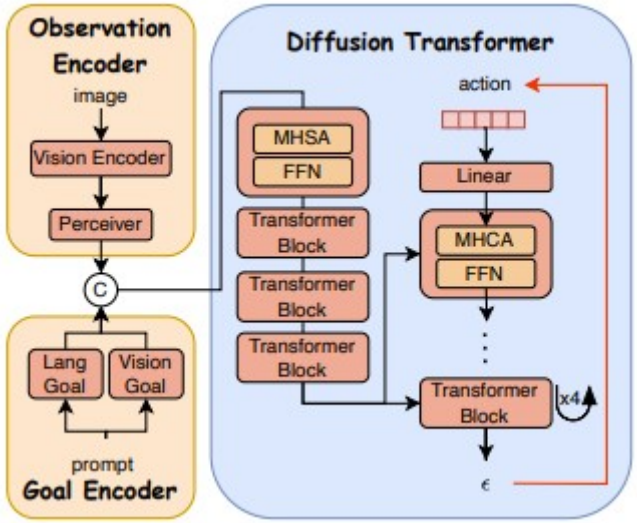
Search 1,243 efficient Video LLMs methods and found 95 representative works

Re-evaluate 21 Open-source codes on 4 complete video benchmarks with both efficiency and accuracy metrics using same hardware protocols and same model backbones

[*Wang et al. ArXiv* 2026] “Rethinking Efficiency in Video LLMs: A Systematic Review and Evaluation" *in ArXiv 2026*

Robotic AI Mo
Efficiency – Mo
Compression

Latency Observation of DP Model



Components	IE	DT
Latency (ms)	1.28	0.906
Parameter (M)	11.2	8.97
NFE	1	100
Total Latency (ms)	1.28	90.6

DP-T Model on iPhone 13

Components	GLE	IE	DT
Latency (ms)	3.74	3.78	2.25
Parameter (M)	151.28	111.05	22.52
NFE	1	2	10
Total Latency (ms)	3.74	7.56	22.25

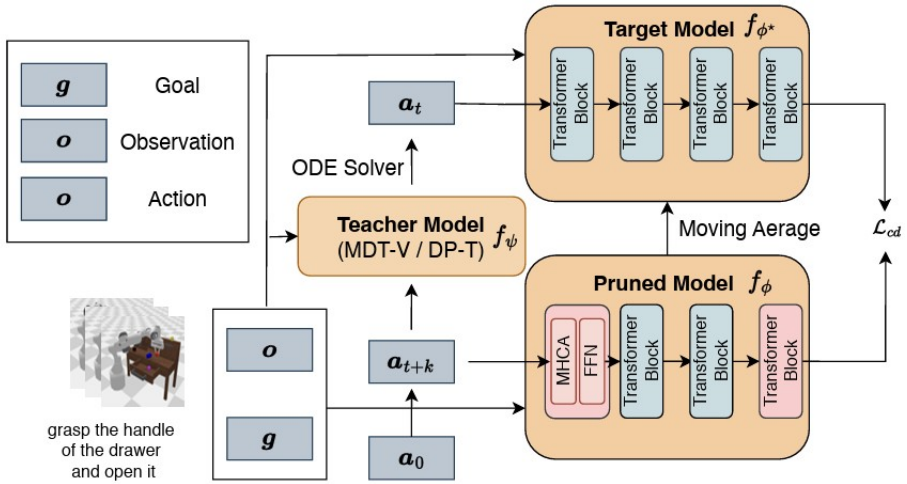
MDT-V Model on iPhone 13

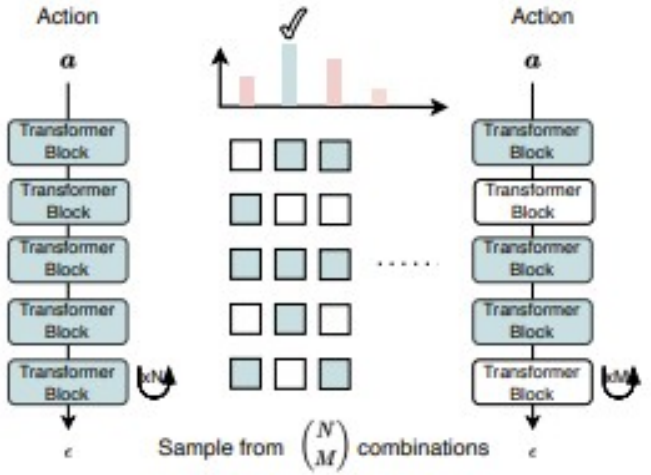
IE: Image Encoder

DT: Diffusion Transformer

GLE: Goal Language Encoder

Our Solution





[*Wu, Wang & Chen, ICCV* 2025] "On-Device Diffusion Transformer Policy for Efficient Robot Manipulation *in IEEE ICCV 2025*

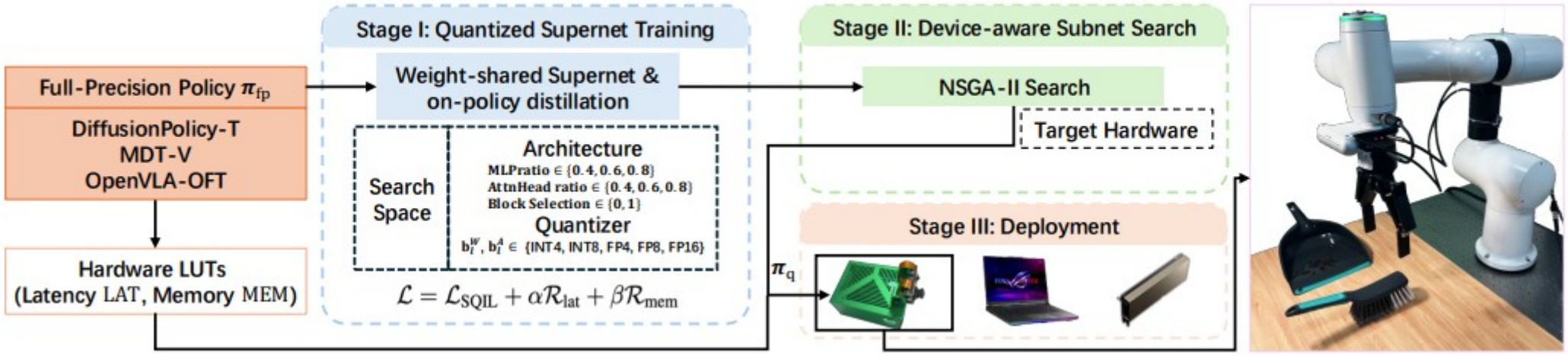
Robotic AI Model Efficiency – Model Compression

Method	Depth	Param (M)	NFE	GFLOPs	Inference Speed (ms)	Success Rate
DP-T DP-T*	8	8.97	100	4.39	90.6	0.772±0.039 0.754±0.023
DP-T-D6/6-8 DP-T-D6/4-4	6	6.87	4	0.134	4.79	0.752±0.019 0.732±0.034
DP-T-D4/4-8 DP-T-D4/2-4 DP-T-D4/1-2	4	4.76	4	0.091	2.72	0.747±0.010 0.732±0.013 0.757±0.018
DP-T-D2/2-8 DP-T-D2/1-4	2	2.65	4	0.049	0.97	0.730±0.022 0.724±0.030



[*Wu, Wang & Chen, ICCV* 2025] "On-Device Diffusion Transformer Policy for Efficient Robot Manipulation *in IEEE ICCV 2025*

Robotic AI Model Efficiency – Train once Compress Model for Every Tasks



We develop a pipeline for Once-for-All Model Compression for Robotics Manipulation

Super-Net: A general Model Compression → device-agnostic quantization & compression
Sub-Net: A specific Model Compression → device-aware compression

[*Wu, Wang & Chen, ArXiv* 2026] "Device-Conditioned Neural Architecture Search for Efficient Robotic Manipulation" *in ArXiv 2026*

Robotic AI Model Efficiency – Train once Compress Model for Every Tasks

First Row is Diffusion Policy based model
Second Row is Visual Language Model based model
We don't need to re-train from DP-based to VLA-based or Task A to Task B

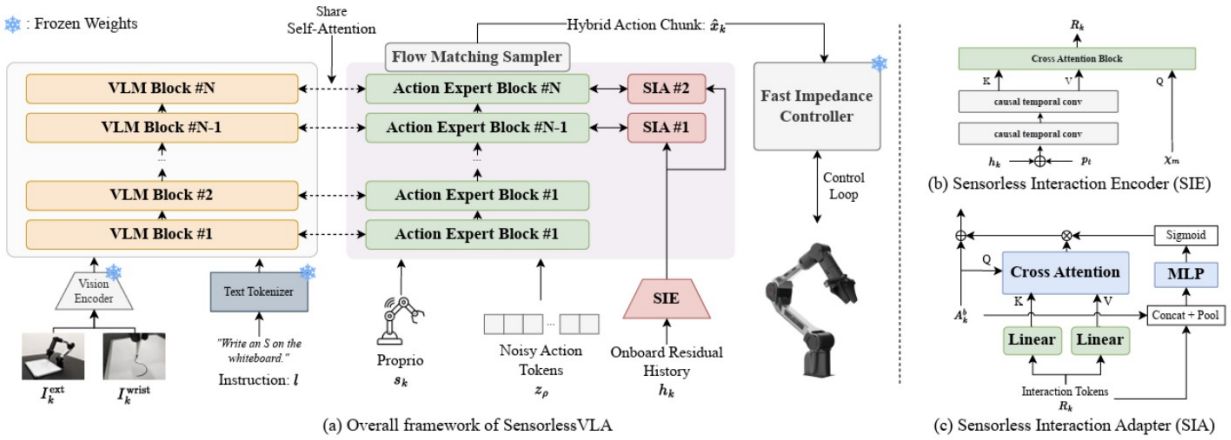
[*Wu, Wang & Chen, ArXiv* 2026] “Device-Conditioned Neural Architecture Search for Efficient Robotic Manipulation" *in ArXiv 2026*

Robotic AI Model Efficiency – - Train once Compress Model for Every Tasks

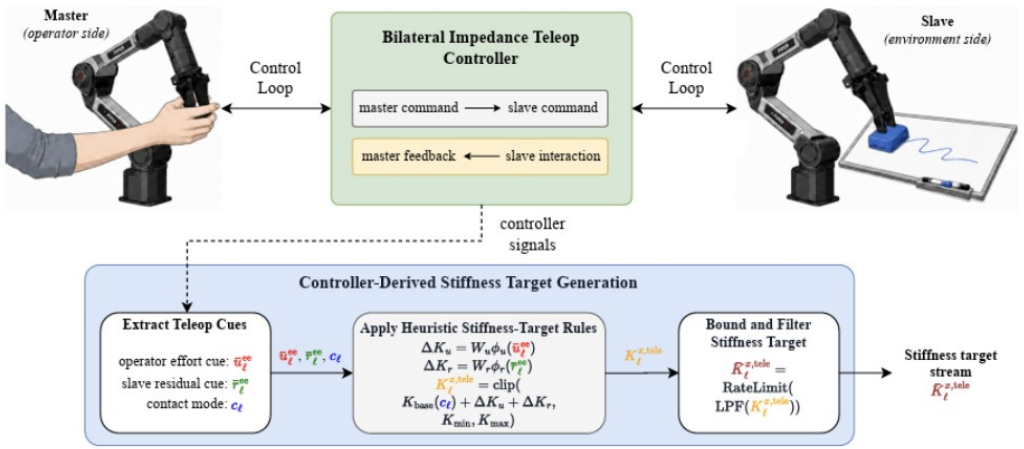
First Row is Diffusion Policy based model
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[*Wu, Wang & Chen, ArXiv* 2026] “Device-Conditioned Neural Architecture Search for Efficient Robotic Manipulation" *in ArXiv 2026*

Robotic AI Model Efficiency – Towards Hardware



A sensorless joint VLA-control framework that couples an interaction-conditioned policy with a standard high-rate impedance controller.



A sensorless joint VLA-control framework that couples an interaction-conditioned policy with a standard high-rate impedance controller.

[*Yang, Chen & Xu, ArXiv* 2026] "SensorlessVLA: A Sensorless Joint VLA-Control Contact-Rich Robotic Manipulation Framework" *in ArXiv 2026*

Robotic AI Model Efficiency – Towards Hardware

TABLE III
MAIN REAL-ROBOT SUCCESS RATES (%, 30 ROLLOUTS PER TASK).

Method	Erase	Write-S	Cable	C-Wipe	Avg.
π_0-Motion [8]	40.0	20.0	26.7	36.7	30.8
$\pi_{0.5}$-Motion [9]	33.3	26.7	30.0	40.0	32.5
SmolVLA-Motion [11]	43.3	16.7	26.7	46.7	33.3
SensorlessVLA-Motion-Only	43.3	40.0	33.3	50.0	41.7
SensorlessVLA-Fixed-Stiffness	66.7	60.0	40.0	53.3	55.0
SensorlessVLA	**80.0**	**73.3**	**60.0**	**70.0**	**70.8**

VS other trajectory-only baselines

TABLE IV
EXTERNAL F/T COMPARISON IN THE SEPARATE SENSOR-MOUNTED CAMPAIGN (%, 30 ROLLOUTS PER TASK).

Condition	Erase	Write-S	Cable	C-Wipe	Avg.
Sensor mounted, input masked	76.7	70.0	63.3	70.0	70.0
External F/T input enabled	**83.3**	**73.3**	63.3	70.0	**72.5**

VS Sensor-based baselines

Thanks
&
Question

